# Exact Local Potentials for the Randić–Radius Problem

The degree-seven case and new randomized quantum mixing families

Keio Research Notes
With internal proof audits and exact computational certificates

2 November 2026

**Abstract**

We prove that every finite connected simple graph of maximum degree at most seven satisfies $\sum_{uv\in E(G)} 1/\max(d_u, d_v) \geq \rho(G)$, where $\rho$ is the minimum eccentricity over vertex and edge-interior centres with demand at the original vertices. Consequently its Randić index is at least $\mathrm{rad}(G) - 1/2$. The proof combines a minimal-counterexample reduction with exact rational local potentials. A first certificate bounds a hypothetical core by twelve vertices; fourteen further dual certificates and one simplicity constraint reduce this to eleven. An exhaustive integer-arithmetic core check completes the proof. The source, certificate data, and verifiers are included in the single TeX file. Independently, we construct bounded compactly supported observation-time densities giving exact uniform quantum mixing on $P_{3p-1}$ for every prime $p \notin \{5, 7\}$; the excluded cases are impossible by an existing obstruction. A second construction covers $P_{2^A p^k - 1}$ for primes $p \geq 5$, every $A \geq 1$, and every $k \geq 2$. These are advances on specified infinite classes. Neither the unrestricted Fajtlowicz conjecture nor the classification of all quantum paths is settled here. Historical priority and comparative research value are not certified.

## 1 Original targets and the exact scope of the results

Fajtlowicz's 1988 conjecture asks whether every connected graph satisfies $R(G) \geq \mathrm{rad}(G) - 1$, where

$$R(G) = \sum_{uv\in E(G)} \frac{1}{\sqrt{d_u d_v}}.$$

We retain this original statement. Our graph theorem proves a stronger inequality on the class $\Delta(G) \leq 7$, with no restriction on the number of vertices. It is a bounded-degree case, not a proof of the unrestricted named conjecture. Cygan, Pilipczuk and Škrekovski proved the corresponding classical radius conclusions for maximum degree four [1]. Doig's later work treats cactus graphs and gives further context [2].

For quantum walks, the target is exact uniformity of all averaged transition probabilities under a single freely chosen observation-time law. This is the randomized observation problem discussed by Coutinho and Guo [5] and studied for paths by Kang [6]. Our family classifications use this same notion. They do not concern uniformity of the standard Cesàro time average, instantaneous uniform mixing, or the separate rank-two average-mixing problem. Published cases and obstructions are distinguished from the new positive constructions below.

**Starting material.** We inspected the 102 numbered research drafts in the repository [3], at commit

`ecd87de3348dcb706a76bb12fde722d7a348a012`.

Its timestamp is 2 November 2026, 19:38:17, UTC+09:00; a fresh fetch confirmed this snapshot. The repository's own provenance statement identifies it as an unrefereed collection and warns that it may contain errors. We use its statements as research leads and reproduce the analytic reduction needed from [087]. The preceding research note's factorial core bound and odd-prime-power mixing theorem are starting points, not results counted again in this round.

**Methodological contribution and limits.** The auxiliary graph problem is to assign rational potentials to local degree types so that every permitted edge pays for the local vertex inequality. Solving this finite problem converts a qualitative minimal-counterexample reduction into a small, exact certificate. The quantum construction instead prescribes moments on the actual frequency-relation torus and then corrects them on a finite interval of observation times. A positive quadratic density supplies correlations with a dependent phase; a separate root-label construction handles a fixed zero mode. These are concrete constructions relative to the inspected source material. Linear programming duality, character orthogonality and harmonic analysis are established tools; we do not assert a historically unprecedented foundational principle.

The requested conjunction of an unchanged previously open named problem, a complete unrestricted solution, historical novelty, and higher value than all previous work has *not* been established. Complete proofs for the stated classes, a bounded literature search, and global priority are different claims. Exact computations and independent audits support the results here; they are not a proof-assistant formalization or external peer review.

## 2 The modified Randić–radius inequality through degree seven

For a finite simple graph put

$$M(G)=\sum_{uv\in E(G)}\frac{1}{\max(d_G(u),d_G(v))},\qquad R(G)=\sum_{uv\in E(G)}\frac{1}{\sqrt{d_G(u)d_G(v)}}.$$

Let $\Gamma(G)$ replace every edge by a unit segment, and define the *vertex-demand absolute radius*

$$\rho(G)=\min_{x\in\Gamma(G)}\max_{v\in V(G)}d(x,v).$$

The maximum here is over graph vertices, not over all points of the metric realization. This distinction matters, for example on cycles.

**Theorem 2.1.** *Every connected finite simple graph of maximum degree at most seven satisfies*

$$M(G)\geq\rho(G)\geq\operatorname{rad}(G)-\tfrac{1}{2}.$$

*Consequently* $R(G)\geq\operatorname{rad}(G)-\frac{1}{2}$*, and in particular Fajtlowicz's inequality* $R(G)\geq\operatorname{rad}(G)-1$ *holds throughout this class.*

This is a bounded-degree result. It does not prove either the unrestricted Fajtlowicz conjecture or the unrestricted Caporossi–Hansen conjecture. For the classical Randić–radius problem, the established chemical-graph case has maximum degree four [1]; corpus manuscript [4] develops the absolute-radius reduction and proves the degree-five case. We reproduce the necessary analytic reduction below. The new step is an exact degree-type potential: it bounds every residual core by twelve vertices, and a second finite certificate excludes order twelve. All rational certificates and their standard-library verifier are included in the accompanying single source. No claim that a literature search establishes worldwide priority is made.

### 2.1 Elementary metric and deletion identities

The minimum defining $\rho$ is attained at a vertex or an edge midpoint. Indeed, on an edge $uv$, at coordinate $t \in [0,1]$, distance to a vertex $w$ is $\min\{t + d(u,w), 1 - t + d(v,w)\}$; the two integer endpoint distances differ by at most one. The upper envelope has a minimum at 0, 1/2, or 1. Thus

$$2\rho(G) = \min\left\{2\,\mathrm{rad}(G),\ \min_{uv\in E(G)}\left(1 + 2\max_{w}\min\{d(u,w), d(v,w)\}\right)\right\}, \tag{1}$$

so $\rho \in \{\mathrm{rad} - 1/2, \mathrm{rad}\}$. Deleting a non-bridge edge cannot decrease $\rho$.

For a leaf $\ell$ with support $p$, $d = d_G(p) \geq 2$, let $\lambda$ count neighbors of $p$ other than $\ell$ whose full degrees are less than $d$. Directly recomputing affected weights gives

$$M(G) - M(G - \ell) = \frac{1}{d} - \frac{\lambda}{d(d-1)} \geq 0. \tag{2}$$

For an edge $uv$ with both degrees at least two, let $\beta_u$ count lower-degree neighbors of $u$ other than $v$, and define $\beta_v$ symmetrically. Then

$$M(G - uv) - M(G) = \frac{\beta_u}{d_u(d_u - 1)} + \frac{\beta_v}{d_v(d_v - 1)} - \frac{1}{\max(d_u, d_v)}. \tag{3}$$

An edge is called *supportive* when this change is positive.

**Lemma 2.2.** *For a connected graph with at least three vertices and a leaf $\ell$,*

$$\rho(G) - \tfrac{1}{2} \leq \rho(G - \ell) \leq \rho(G).$$

*Deleting any set of leaves lowers $\rho$ by at most one. If two leaves have the same support and $G$ is not a star, deleting one of them leaves $\rho$ unchanged.*

*Proof.* Distances between retained vertices and edge points do not change. A center lying in a deleted pendant edge can be moved to its support, proving the upper bound. Let $y$ be a vertex or midpoint center of $G - \ell$, with value $\rho'$, and let $p$ support $\ell$. If $d(y,p) \leq \rho' - 1/2$, the same point has value at most $\rho' + 1/2$ in $G$. Otherwise $d(y,p) = \rho'$, since both quantities are half-integers. Move $y$ by $1/2$ toward $p$. All retained vertices are then within $\rho' + 1/2$, as is $\ell$. This proves the lower bound. For a set of leaves, a center of the retained graph is within one extra unit of every removed leaf. Finally, for two leaves at $p$, choose a center of the graph after one deletion outside the remaining pendant edge: if necessary, its projection to $p$ is at least as good. The remaining leaf then has the same distance as the deleted one, proving equality. □

**Lemma 2.3** (Cut-vertex padding)**.** *Suppose $v$ is a cut vertex and $V(G) \setminus \{v\} = A \sqcup B$ with no edge from $A$ to $B$, where both $G[A]$ and $G[B]$ contain an edge. Let $H_A = G[A \cup \{v\}]$, and obtain $G_A$ by adding $d_B(v)$ new leaves at $v$. Define $G_B$ symmetrically. Then degrees of retained vertices are preserved, and*

$$M(G) = M(G_A) + M(G_B) - 1, \qquad \rho(G) \leq \rho(G_A) + \rho(G_B) - 1.$$

*Proof.* The two padded graphs preserve all original edge weights, while their new leaf weights sum to $(d_A(v) + d_B(v))/d_G(v) = 1$. Put $\rho_A = \rho(G_A)$ and choose a center $y_A \in \Gamma(H_A)$; a center in a new pendant edge can be projected to $v$. Set $s_A = d(y_A, v)$. Since $G_A$ has a leaf at $v$, $s_A \leq \rho_A - 1$. Define $y_B, s_B$ similarly. On a geodesic from $y_A$ through $v$ to $y_B$, with signed coordinate $p \in [-s_A, s_B]$, every vertex is within

$$\max\{\rho_A + s_A + p, \rho_B + s_B - p\}.$$

If the balancing point lies in this interval, the minimum of this upper bound is $(\rho_A + \rho_B + s_A + s_B)/2 \leq \rho_A + \rho_B - 1$. If it lies outside, one endpoint gives at most $\max(\rho_A, \rho_B)$, which is also at most $\rho_A + \rho_B - 1$ because both radii are at least one. □

## 2.2 The residual core and the source budget

**Proposition 2.4.** *If $M \geq \rho$ fails among connected graphs of maximum degree at most $D \geq 3$, a counterexample exists of the form $G = K + L$, where $K$ is 2-connected of order at least three and $L$ consists of at most one leaf per core vertex, such that:*

*(i) all full core degrees lie in $[3, D]$, every core edge is supportive, and every core vertex has a lower-degree neighbor in $G$;*

*(ii) $\delta := M(G) - M(K) < 1$;*

*(iii) exactly $s \leq 2$ core vertices have core degree two. They have full degree three, carry leaves, are independent, and have core neighbors of full degree at least four.*

*Proof.* Choose a counterexample minimizing lexicographically its cyclomatic number $|E| - |V| + 1$ and then its number of edges. Every non-bridge edge is supportive; otherwise deleting it decreases cyclomatic number, does not increase $M$, and does not decrease $\rho$.

At a cut vertex, at most one component after deletion can contain an edge. Otherwise apply Lemma 2.3. Each padded graph has the same degree bound and smaller lexicographic size: its cyclomatic number cannot increase, and if unchanged, the omitted component's internal edge makes the edge count smaller. Minimality and the two padding inequalities contradict $M(G) < \rho(G)$.

No leaf can have a degree-two support, except in the already valid star $P_3$: deleting such a leaf drops $M$ by exactly $1/2$ and drops $\rho$ by at most $1/2$. No vertex carries two leaves, because deleting one does not increase $M$ and, outside stars, leaves $\rho$ unchanged. Stars satisfy $M \geq \rho$. If $G$ were a tree, the cut-vertex property would make it a star or double star; the preceding restrictions leave only $P_4$ among the latter, already excluded by the degree-two support argument. Thus $G$ contains a 2-connected block $K$ of order at least three. The cut-vertex property forces every vertex outside $K$ to be an attached leaf, proving the asserted form.

A core vertex $u$ with no lower-degree neighbor would have no leaf. For a neighbor $v$ one has $d_v \geq d_u$, and (3), with $\beta_u = 0$ and $\beta_v \leq d_v - 1$, contradicts supportiveness. A full core degree of two is consequently impossible. A full degree-three vertex has to carry a leaf, since all core degrees are at least three; two such vertices cannot be adjacent because (3) would then give zero. Conversely a core-degree-two vertex must carry its one leaf and have full degree three.

Deleting any nonempty subset $L'$ of leaves produces a smaller graph, so minimality and Lemma 2.2 give $M(G) - M(G - L') < \rho(G) - \rho(G - L') \leq 1$. In particular $\delta < 1$. If $L'$ consists of the leaves at the full degree-three vertices, its exact drop is $s/3$: their other neighbors retain degrees at least four. Hence $s/3 < 1$, so $s \leq 2$. □

A *source* is a core vertex without a lower-degree core neighbor. Every source has a leaf. Sources are independent: unequal degrees contradict the definition at the higher endpoint, and for equal degree $d \geq 3$ the deletion change is $(3 - d)/[d(d - 1)] \leq 0$. Deleting just their leaves changes no retained edge weight. Therefore

$$\sum_{v \text{ a source}} \frac{1}{d_v} \leq \delta < 1. \tag{4}$$

In particular, if $N_d$ counts full core degrees and $d_0$ is the least occupied degree, then $N_{d_0} < d_0$. This budget will remove degree profiles before the final finite certificate.

## 2.3 Degree-type potentials and the order-twelve exclusion

In this subsection, a decorated core has minimum core degree at least two and at most one added leaf at each core vertex; it is supportive if every core edge is supportive. For a core vertex define

its type $t = (d, p, l)$ by its full degree, leaf indicator, and number of lower-degree neighbors in $G$. Put $g_d = d(d-1)$. For $D = 7$ all possible types belong to the following set of size 45:

$$\mathcal{T} = \{(3,1,1)\} \cup \{(d,p,l) : 4 \le d \le 7,\ p \in \{0,1\},\ 1 \le l \le d\}.$$

Including types that cannot actually occur only enlarges our relaxation. For an edge joining $t = (d, p, l)$ to $u = (e, q, m)$ with $d < e$, supportiveness is equivalent to

$$\frac{l}{g_d} > \frac{e-m}{g_e}; \qquad \text{for } d = e \text{ it is equivalent to } l + m \ge d. \tag{5}$$

These tests leave 802 allowed unordered type pairs, including pairs of the same type. A vertex of type $t$ has $l - p$ lower-degree core neighbors and $d - l$ equal-or-higher-degree core neighbors.

Simultaneously deleting all leaves gives the exact identity

$$\delta = \sum_t n_t \frac{p(d-l)}{g_d} - \sum_{\substack{uv \in E(K) \\ d_u = d_v = d,\ p_u = p_v = 1}} \frac{1}{g_d}. \tag{6}$$

Indeed, deleting a host's leaf removes $1/d$ and increases each edge to a lower-degree retained neighbor by $1/g_d$. Equal-degree edges increase only when both endpoints are hosts, producing the final correction.

**Lemma 2.5** (Rational potential certificate)**.** *Every supportive decorated core with full degrees in* $[3, 7]$ *satisfies*

$$|V(K)| \le \frac{875}{68}\,\delta.$$

*Proof.* The included certificate assigns rational numbers $\alpha_t, \beta_t$ to the 45 types. With $C = 875/68$, it verifies exactly

$$(l-p)\alpha_t + (d-l)\beta_t + C\frac{p(d-l)}{g_d} \ge 1, \tag{7}$$

$$\beta_t + \alpha_u \le 0 \qquad (d < e), \tag{8}$$

$$\beta_t + \beta_u + C\frac{pq}{g_d} \le 0 \qquad (d = e), \tag{9}$$

where the last two inequalities range over allowed pairs only. These are $45 + 802$ explicit rational comparisons. Sum (7) over vertices. Each unequal-degree edge contributes $\beta$ at its lower endpoint and $\alpha$ at its higher endpoint; an equal-degree edge contributes two $\beta$ terms. Applying (8)–(9) and (6) proves the lemma. The supplied verifier checks the certificate with integer fractions and does not invoke an optimizer. □

Thus $\delta < 1$ forces $|K| \le 12$. To exclude twelve, let $N = (N_3, N_4, N_5, N_6, N_7)$. The exact candidate list consists of

$$N_d \ge 0, \quad \sum N_d = 12, \quad N_3 \le 2, \quad N_{d_0} < d_0 \text{ for the least occupied degree } d_0. \tag{10}$$

There are 896 such integer profiles. At a fixed profile, discard any type failing

$$N_d > 0, \qquad l - p \le \sum_{e<d} N_e, \qquad d - l \le \sum_{e \ge d} N_e - 1. \tag{11}$$

Every discarded type violates a necessary distinct-neighbor count.

**Lemma 2.6** (Finite profile exclusion)**.** *No supportive decorated core of order twelve, with full degrees in* $[3, 7]$ *and* $\delta < 1$*, exists.*

*Proof.* We specify the finite certificate mathematically so that its checks are independent of how its coefficients were found. Let $x_{tu}$ be the number of allowed core edges of type pair $tu$. The lower and upper incidence equations at type $t = (d, p, l)$ are

$$\sum_{u:d_u<d} x_{tu} = (l-p)n_t, \qquad \sum_{u:d_u\geq d} x_{tu} = (d-l)n_t,$$

where a same-type edge contributes twice to the second sum. Add the five degree-count equations $\sum_{t:d_t=d} n_t = N_d$. The objective is the right side of (6).

A dual vector comprises two unrestricted rational multipliers per type and five degree-count multipliers. It is a certificate of a lower bound on $\delta$ if its coefficient on every retained vertex or edge variable is at most that variable's coefficient in (6). The resulting bound is the dot product of its last five entries with $N$. All variables being nonnegative, this implication follows by summing inequalities, without linear-programming duality as an external assumption.

The included data contain 14 such rational vectors. The verifier reconstructs all profiles (10), retains the columns specified by (11) and (5), and confirms that at least one vector proves $\delta \geq 1$ for each of 895 profiles. Its sole residual profile is

$$N = (0, 3, 3, 3, 3).$$

For this profile let $Z$ be the degree-six vertices with $l = 1$. They form an independent set by (5). Every one needs five distinct equal-or-higher-degree neighbors. There are only six vertices of degree at least six, so $|Z| \leq 1$ (the case $Z$ empty is immediate). Incorporating this one additional inequality, the residual rational certificate proves

$$\delta \geq \frac{103}{84} > 1.$$

Its multiplier on $|Z| \leq 1$ is $-61/420$; all other added capacity multipliers are zero. The verifier checks every coefficient and the exact lower bound. This excludes the residual case and proves the lemma. □

The certificate data are small: one 45-type potential, fourteen 95-entry profile vectors, and one residual vector. The single source contains their compressed numerical payload and a Python verifier using only `fractions`, `json`, and other standard-library modules. The reconstructed profile count, uncovered profile, and exact residual bound are all asserted by the verifier. No floating-point solver result is used as a proof certificate.

### 2.4 The finite core inequality and conclusion

**Lemma 2.7** (Exhaustive finite core certificate)**.** *If $K$ is 2-connected, has at most eleven vertices, maximum degree at most seven, minimum degree at least two, and exactly $s \leq 2$ vertices of degree two, then*

$$\rho(K) \leq M(K) - 1 + \frac{s}{3}. \tag{12}$$

*Proof.* Generate every 2-connected graph of the stated order and degree bounds with `geng`; retain precisely those with $s \leq 2$. For each retained graph, compute all distances by breadth-first search and $2\rho$ by (1). Let $W = \operatorname{lcm}(6, 2, 3, \ldots, \Delta(K))$ and let $J = WM(K)$, computed as a sum of integers $W/\max(d_u, d_v)$. The exact integer comparison is

$$3W(2\rho) \leq 6J - 6W + 2Ws.$$

No violation occurs. The enumeration and integer checker are included in the reproducibility material. The counts below distinguish the unrestricted-degree checks through order nine from the degree-seven checks at orders ten and eleven; the former are a safe enlargement.

| Order | Maximum-degree filter | Eligible cores | Minimum slack |
|---|---|---|---|
| 4 | none | 2 | 0 |
| 5 | none | 6 | 1/6 |
| 6 | none | 44 | 1/6 |
| 7 | none | 385 | 1/4 |
| 8 | none | 6 268 | 3/10 |
| 9 | none | 179 303 | 1/3 |
| 10 | 7 | 7 355 341 | 5/14 |
| 11 | 7 | 473 655 792 | 1/2 |

Slack means $M(K) - 1 + s/3 - \rho(K)$. Order three has no eligible graph. An independent Python implementation, using `Fraction` and separate graph decoding and distance calculations, reproduced every count and slack through order nine, for 186 008 eligible cores. The higher-order C run was newly executed, rather than inferred from the earlier corpus table. Enumeration completeness uses the standard `nauty` generator; this is a computer-assisted finite proof, not a proof-assistant formalization. □

*Proof of Theorem 2.1.* If a counterexample of maximum degree seven exists, Proposition 2.4 supplies $G = K + L$ with $\delta < 1$. Lemmas 2.5 and 2.6 give $|K| \le 11$. Let $L_3$ be the leaves attached at the $s$ full degree-three core vertices. Their exact deletion drop is $s/3$, and subsequent leaf deletions do not increase $M$. Thus $M(K) \le M(G) - s/3$. Adding leaves raises $\rho$ by at most one. By Lemma 2.7,

$$\rho(G) \le \rho(K) + 1 \le M(K) + s/3 \le M(G),$$

a contradiction. Finally, $R \ge M$ termwise and $\rho \ge \mathrm{rad}\, -1/2$ by (1). □

For reproducibility, the generator was built from the `lonnen/nauty` source mirror, release marker `This_is_nauty2_9_0.txt`, commit `3915d891af81630853ab485f2696828a31253cd9`. The high-order commands are

```
geng -C -d2 -D7 -q 10 | graph_corechk
geng -C -d2 -D7 -q 11 | graph_corechk
python3 graph_verify_compact.py graph_degree7_compact_certificate.json
```

The generator, independent rational certificate, and two graph checkers serve different roles: generation establishes finite coverage, the rational certificate proves the reduction in order, and the checkers evaluate the finite core inequality. The bounded-degree restriction remains essential to the result stated here.

## 3 Randomized uniform mixing when the path modulus is three times a prime

For a graph with adjacency matrix $A$, a probability law for the observation time $T \ge 0$ gives *randomized uniform mixing* if

$$\mathbb{E}|(e^{iTA})_{uv}|^2 = \frac{1}{|V|} \qquad (u, v \in V).$$

The same law must work for every initial vertex. This is the ordinary adjacency quantum walk with a freely chosen time law. It is different from the standard Cesàro average, which fixes the averaging protocol to $\lim_{L\to\infty} L^{-1} \int_0^L$. In particular, nothing here settles the rank-two Cesàro question in [5, Open Problem 4.1]. The randomized path question is discussed separately in [5, Section 4.2].

**Theorem 3.1** (The three-times-prime family)**.** *Let $p$ be a prime. The unweighted path $P_{3p-1}$ admits randomized uniform mixing if and only if $p \notin \{5, 7\}$. In every positive case, the time law can be chosen to have a bounded, compactly supported probability density on $[0, \infty)$.*

Kang's negative theorem already excludes $p = 5, 7$, and his positive theorem includes $p = 2$ [6]. The additional cases established below are the surviving members of this particular arithmetic family. This is not an all-path classification.

### 3.1 Spectral target and exact realization in time

Write $q = n + 1$ for the modulus of $P_n$. Its eigenvalues, normalized eigenvectors and spectral projectors are

$$\lambda_j = 2\cos(\pi j/q), \qquad v_j(u) = \sqrt{2/q}\sin(\pi ju/q), \qquad E_j = v_j v_j^{\mathsf{T}} \quad (1 \le j, u < q).$$

Let $R$ reverse the vertex order and let $\circ$ denote the entrywise product. Two useful identities are

$$D_0 := \sum_j E_j \circ E_j = \frac{1}{q}\left(J + \frac{I + R}{2}\right), \qquad \sum_{j \equiv \ell \pmod 2} E_j \circ E_\ell = \frac{I + R}{2}. \tag{13}$$

For the first identity, expand the product of two squared sines and use $\sum_{j=1}^{q-1}\cos(2\pi aj/q) = q\mathbf{1}_{q|a} - 1$. For the second, use $Rv_j = (-1)^{j+1}v_j$ and expand

$$\frac{1}{2}\left[\left(\sum_j E_j\right) \circ \left(\sum_j E_j\right) + \left(\sum_j (-1)^j E_j\right) \circ \left(\sum_j (-1)^j E_j\right)\right].$$

Consequently, phase coherences equal to 1 on the diagonal, $-c$ on distinct equal-parity pairs, and 0 on opposite-parity pairs give the mixing matrix

$$(1 + c)D_0 - c\frac{I + R}{2} = \frac{J}{n} \qquad \text{when } c = 1/n. \tag{14}$$

Only real parts of these coherences enter the mixing matrix, although our constructions satisfy the displayed complex moments as well.

When $q$ is odd, each pair $j, q - j$ has one odd and one even member, and $\lambda_{q-j} = -\lambda_j$. Thus it suffices to construct the $m = (q - 1)/2$ odd-index phases with

$$\mathbb{E}W_i\overline{W}_j = -c \quad (i \ne j), \qquad \mathbb{E}W_iW_j = 0 \quad (\text{all } i, j), \tag{15}$$

and take even phases to be their reflected conjugates. All these laws must be supported on the actual orbit closure of the time flow.

The next lemma supplies the passage from that closure to an actual observation-time law. It will also be used for other path families.

**Lemma 3.2** (Exact finite-time realization)**.** *Let $\mathcal{T}$ be a finite-dimensional torus, and let $\gamma(t)$, $t \ge 0$, be an equidistributed linear flow in $\mathcal{T}$. Let $F_1, \dots, F_D$ be distinct nontrivial cosine characters, identified modulo sign. Suppose a probability measure $\mu$ on $\mathcal{T}$ has the desired finite moment vector $b_j = \int F_j\, d\mu$ and satisfies $\mu \ge \eta H$ for some $\eta > 0$, where $H$ is Haar probability. Then a bounded, compactly supported probability density $\rho$ on $[0, \infty)$ satisfies $\int_0^\infty F_j(\gamma(t))\rho(t)\, dt = b_j$ for every $j$.*

*Proof.* Inversion-symmetrize $\mu$, which preserves its cosine moments and its Haar lower bound. Convolve it with a nonnegative product Fejér approximate identity. The resulting trigonometric-polynomial density is at least $\eta$, and its finitely many specified moments converge to $b_j$. Distinct

cosine characters have Haar inner products $\int F_i F_j \, dH = \delta_{ij}/2$. If the moment error is $\delta_j$, adding $2\sum_j \delta_j F_j$ corrects every moment, preserves the integral, and changes the sup norm by at most $2\sum_j |\delta_j|$. Choosing the approximation sufficiently close gives a trigonometric polynomial $f \geq \eta/2$ with exactly the desired moments.

Put $F_0 = 1$, $b_0 = 1$, and regard $F$ and $b$ as column vectors. Along the flow define

$$G_L = \frac{1}{L}\int_0^L F(\gamma(t))F(\gamma(t))^{\mathsf{T}}\,dt, \qquad b_L = \frac{1}{L}\int_0^L F(\gamma(t))f(\gamma(t))\,dt.$$

Equidistribution gives $G_L \to \operatorname{diag}(1, 1/2, \ldots, 1/2)$ and $b_L \to b$. Therefore, for all sufficiently large finite $L$, $a_L = G_L^{-1}(b - b_L)$ exists and tends to zero. Set

$$\rho(t) = \frac{f(\gamma(t)) + a_L^{\mathsf{T}} F(\gamma(t))}{L}\mathbf{1}_{[0,L]}(t).$$

For sufficiently large $L$ this is nonnegative, since $f \geq \eta/2$. The defining linear system gives $\int \rho F = b$, including $\int \rho = 1$. Boundedness and compact support are immediate. □

For the path constructions, a law satisfying (15) at some $c > c_0 = 1/n$ supplies the lemma's Haar component by taking

$$\mu = (1 - c_0/c)H + (c_0/c)\nu_c. \tag{16}$$

Every relevant off-diagonal Hermitian or holomorphic character has zero Haar integral: distinct odd eigenvalues are neither equal nor opposite, and none is zero. The mixture therefore has exactly the target moments. For the full spectrum, the analogous assertion uses the nonzero eigenvalue differences. Repeated cosine characters are listed just once when applying Lemma 3.2.

### 3.2 The exact relation torus for $q = 3p$

Assume $p \geq 5$, and put

$$r = \frac{p-1}{2}, \qquad m = 3r + 1, \qquad n = 6r + 2.$$

The odd indices partition into the singleton $p$ and the $r$ triples

$$\{j, 2p - j, 2p + j\}, \qquad j = 1, 3, \ldots, p - 2.$$

Each triple of frequencies sums to zero. Exactly one index in each triple is divisible by 3; call its phase $U_a$. The sum of those frequencies is

$$\sum_{b=1,3,\ldots,p-2} 2\cos(\pi b/p) = 1 = \lambda_p. \tag{17}$$

The trigonometric sum is $\sin((p-1)\pi/p)/\sin(\pi/p) = 1$. Hence the singleton phase satisfies $W_0 = \prod_{a=1}^{r} U_a$.

These $r + 1$ relations generate the entire integer relation lattice. Indeed, the real cyclotomic field $K = \mathbb{Q}(2\cos(\pi/q))$ has degree $\varphi(6p)/2 = p - 1 = 2r$. The odd-index eigenvalues span $K$: powers of $\lambda_1$ expand into 1 and frequencies $2\cos(a\pi/q)$, periodicity and reflection reduce all these to the odd modes, and $1 = \lambda_p$ is already present. Thus their rational relation space has dimension $m - 2r = r + 1$. The disjoint triple relations and (17) are independent, because only the latter has a nonzero singleton coordinate. To check saturation, take an integral vector in their rational span. Its singleton coordinate forces the coefficient of the extra relation to be integral. After subtracting that multiple, the three coordinates in each disjoint triple force its remaining coefficient to be integral.

The character criterion for a linear torus flow now shows that its closure is exactly $\mathcal{T} \cong (S^1)^{2r}$ with characters

$$\chi_{a1} = u_a, \qquad \chi_{a2} = v_a, \qquad \chi_{a3} = -u_a - v_a, \qquad \gamma = \sum_{a=1}^{r} u_a. \tag{18}$$

Here $u_a, v_a$ are a free integral character basis, $U_a$ is the phase of $u_a$, and $W_0$ is the phase of $\gamma$. Every nontrivial character of this torus has a nonzero time frequency; its time average tends to zero. This also proves the equidistribution needed in Lemma 3.2.

### 3.3 Anticorrelating the triple modes while cancelling singleton moments

Call the $3r$ triple phases the old modes. For $r \geq 4$, we first construct a law $\nu_d$ with old-mode off-diagonal Hermitian moments $-d$, all holomorphic moments zero, and all singleton–old Hermitian moments zero. The allowable interval is

$$0 < d \leq d_{\max} := \frac{r-2}{3r^2 - r - 4}. \tag{19}$$

Put $B = \sqrt{3 - 6d}$ and choose $\theta$ with $1 + 2\cos\theta = B$. For each triple use an independently uniform permutation of

$$Y_a(1, e^{i\theta}, e^{-i\theta}), \qquad Y_a^3 = 1. \tag{20}$$

The designated $U_a$ is its fixed indexed coordinate, and the product of the three entries is one. It remains to construct exchangeable labels with pair correlation

$$\mathbb{E} Y_a \overline{Y}_b = -A_d, \qquad A_d = \frac{3d}{1-2d} \quad (a \neq b).$$

Choose positive $s_1 + s_2 = r$ with neither size divisible by 3. One can take $(1, r-1)$ if $r \equiv 0, 2 \pmod 3$, and $(2, r-2)$ if $r \equiv 1 \pmod 3$. Uniformly partition the $r$ positions into these two sizes. Within each group distribute labels among the three cube roots with counts differing by at most one, uniformly permute, and apply an independent common uniform cube-root rotation. In a group of size $s \not\equiv 0 \pmod 3$, the label sum has squared modulus one. Its ordered off-diagonal correlation sum is therefore $-(s-1)$. The groups have zero cross correlations, so exchangeability makes the overall off-diagonal label correlation

$$-A_{\max} = -\frac{r-2}{r(r-1)}.$$

Mix this distribution with independent uniform labels to attain any $0 \leq A_d \leq A_{\max}$, which is precisely (19). Both component laws may be generated with the same two independent group rotations.

The within-triple Hermitian correlation in (20) is $(B^2 - 3)/6 = -d$. Between different triples it is $-B^2 A_d/9 = -d$. All old-mode holomorphic second moments vanish: use a group's common rotation, of charge two if both factors are in that group and of charge one otherwise.

The global relation $W_0 = \prod U_a$ is now useful. Under the common rotation $\omega$ of a group of size $s$, it changes by $\omega^s$. For a singleton–old Hermitian or holomorphic moment, rotate the group not containing the old phase. Since $3 \nmid s$, the moment vanishes. For $\mathbb{E} W_0^2$, either group has nonzero charge $2s$ modulo 3, so that moment also vanishes. This proves all the asserted properties of $\nu_d$ on the exact torus (18).

### 3.4 A positive density supplies the missing singleton correlations

Let $S$ be the sum of the $L = 3r$ old phases. For $\alpha > 0$ define on Haar measure $H$ of the actual torus

$$f_\alpha = \frac{|W_0 - \alpha S|^2}{Z}, \qquad Z = 1 + L\alpha^2. \tag{21}$$

For $r \geq 4$, its moments are

$$\mathbb{E}_{f_\alpha H} W_i \overline{W}_j = \frac{\alpha^2}{Z} \ (i \neq j \text{ old}), \qquad \mathbb{E}_{f_\alpha H} W_0 \overline{W}_i = -\frac{\alpha}{Z}, \qquad \mathbb{E}_{f_\alpha H} W_i W_j = 0 \text{ for all modes.} \tag{22}$$

Here is the character check, including possible collisions. In each block the old characters are $u, v, -u - v$; their six nonzero ordered differences are distinct. Differences supported on two different blocks are likewise distinguished by their two nonzero components. Thus all nonzero ordered old differences are distinct. No old sum (including a doubled character) equals an old difference: simultaneous multiplication of every triple by a cube root gives charge two to a sum and charge zero to a difference. No old characters are negatives of one another. The character $\gamma$ in (18) has support on all $r$ blocks, and $\gamma \pm \chi_i$ has support on at least $r - 1$. For $r \geq 4$ these cannot equal old sums or differences, supported on at most two blocks. Moreover, $2\gamma$ cannot be a sum or difference of two old characters. These observations rule out all unintended collisions among the constant term, old differences, and $\pm(\gamma - \chi_i)$ in the expansion of (21). They also rule out every holomorphic second-moment character from this Fourier support. Character orthogonality therefore proves normalization and (22).

Mix $f_\alpha H$ and $\nu_d$ with weights $t$ and $1 - t$. To obtain the common Hermitian value $-c$, set

$$t = cZ/\alpha, \qquad d = \frac{c(1+\alpha)}{1-t}. \tag{23}$$

Then the singleton–old value is $-c$, the old–old value is $-(1-t)d + c\alpha = -c$, and all holomorphic moments remain zero.

Take $\alpha = 1/4$ and $c_0 = 1/(6r + 2)$. At $c = c_0$ the parameters are

$$t_0 = \frac{3r + 16}{24r + 8} < 1, \qquad d_0 = \frac{5}{21r - 8}.$$

For $r \geq 7$ we have $d_0 < d_{\max}$, since

$$(r - 2)(21r - 8) - 5(3r^2 - r - 4) = 6r^2 - 45r + 36 > 0.$$

Explicitly define

$$c_{\max} = \frac{4(r-2)}{18r^2 + 5r - 52}, \qquad c = \frac{c_0 + c_{\max}}{2}. \tag{24}$$

The same inequality gives $c_{\max} > c_0$. Substitution in (23) shows $0 < t < 1$ and $0 < d < d_{\max}$: indeed $c_{\max}$ is exactly the value for $d = d_{\max}$, and its denominator is positive before $t$ reaches one. We have therefore constructed (15) at a strict value $c > c_0$. The Haar mixture (16), Lemma 3.2, and (14) prove the theorem for every prime $p \geq 17$.

### 3.5 Small positive cases and the two obstructions

For $p = 11, 13$, the complete rational certificates in Section 3.6 give laws on the actual phase torus at $c = 1/17 > 1/32$ and $c = 1/20 > 1/38$, respectively. In each case the law is explicitly symmetrized under complex conjugation; thus the exact cosine-moment verification also establishes the stated complex moments. The same Haar mixture and finite-time lemma apply.

For $p = 3$, $q = 9$, the odd frequencies at indices $1, 3, 5, 7$ have the single relation $\lambda_1 + \lambda_5 + \lambda_7 = 0$; the singleton is $\lambda_3 = 1$. This is the entire integral relation lattice, by the same field-degree and

saturation argument (the field has degree three). Use $c = 1/6 > 1/8$, $B = \sqrt{2}$, and uniformly permute $(1, e^{i\theta}, e^{-i\theta})$ in the triple, with $1 + 2\cos\theta = B$. Independently take the singleton in $\{1, -1\}$ with mean $-1/(2\sqrt{2})$, and apply a common uniform cube-root rotation to all four phases. The triple product remains one. The within-triple correlation is $(B^2 - 3)/6 = -1/6$, the singleton–triple correlation is $[-1/(2\sqrt{2})]B/3 = -1/6$, and all holomorphic moments vanish under the common rotation. Haar mixing and the lemma prove the required bounded compactly supported density.

For $p = 2$, $q = 6$, the spectrum is $\sqrt{3}, 1, 0, -1, -\sqrt{3}$ and the two positive frequencies are rationally independent. On its two-torus take the phase ramp $z_j = e^{i(3-j)\xi}$, $1 \le j \le 5$, and give $\xi \in \mathbb{R}/(2\pi\mathbb{Z})$ the density

$$1 - 2c(\cos 2\xi + \cos 4\xi), \qquad c = 9/40 > 1/5,$$

relative to normalized Haar measure. This density is at least $1/10$, and its coherences are $-c$ at nonzero even index differences and zero at odd differences. It respects the spectral pairing and fixed zero mode. Haar mixing on the actual two-torus and Lemma 3.2 finish this case; this is the small instance of Kang's phase-ramp construction [6].

Finally, $p = 5$ and $p = 7$ give moduli 15 and 21, respectively. Kang's forced-coherence theorem and parity-collision obstruction [6, Theorems S13 and S18] exclude every time probability law. Concretely, at $q = 15$, $\lambda_1 - \lambda_3 = \lambda_4 - \lambda_5$, and at $q = 21$, $\lambda_3 - \lambda_6 = \lambda_7 - \lambda_9$. In either identity one pair has equal parity and the other opposite parity, requiring the same cosine moment to equal both $-1/(q-1)$ and zero. This completes Theorem 3.1.

**Scope and priority.** The added proof combines two independently rotated label groups with the nonnegative density (21) to accommodate the extra global frequency relation. Torus approximation, spectral coherences and moment correction have prior antecedents; they are not presented as unprecedented general principles. A targeted review of the primary sources available through 2 November 2026 found no earlier identical $q = 3p$ classification. This is a bounded literature assessment, not a proof of universal absence. The classification of randomized uniform mixing for all paths remains unresolved here.

### 3.6 Exact certificates at $q = 33$ and $q = 39$

For completeness, the following finite certificates supply the two exceptional positive cases used in the $q = 3p$ theorem. They use only rational and integer arithmetic, and all their data are included below.

Let $\alpha = 2\cos(\pi/q)$ and let $P$ be its monic minimal polynomial of degree $D = \varphi(2q)/2$. Each digit string $(z_0, \dots, z_{D-1})$ assigns $\exp(2\pi i z_a/3)$ to the phase of $\alpha^a$. For an odd eigenvalue $\lambda_j = f_j(\alpha)$, with $f_j$ reduced modulo $P$, the corresponding phase is $W_j = \exp(2\pi i \sum_a [f_j]_a z_a/3)$. Each string is given its displayed integer numerator divided by the common denominator. Independently apply a uniform Galois permutation $j \mapsto \min(aj \bmod 2q,\ 2q - (aj \bmod 2q))$, with $1 \le a < q$ and $\gcd(a, 2q) = 1$. Finally symmetrize under complex conjugation.

The exact moments are

$$\mathbb{E}(W_i \overline{W}_j) = -c \quad (i \ne j), \qquad \mathbb{E}(W_i W_j) = 0 \quad (\text{all } i, j),$$

with $c = 1/17 > 1/32$ for $q = 33$ and $c = 1/20 > 1/38$ for $q = 39$. The weights are strictly positive. Mixing with Haar measure and applying the finite-time realization lemma proves exact randomized uniform mixing for $P_{32}$ and $P_{38}$.

The following self-contained Python program verifies every original moment, not just a symmetry-reduced linear system. It uses only the Python standard library. It independently

reconstructs $\Phi_{2q}$, checks $z^D P(z+z^{-1}) = \Phi_{2q}(z)$, reconstructs all eigenvalue polynomials by $f_0 = 2$, $f_1 = x$, $f_j = xf_{j-1} - f_{j-2}$, and checks all moments using $2\cos(2\pi a/3) \in \{2, -1\}$. The two successful outputs are exact integer identities, with no floating-point tolerance.

```
"""Exact self-contained certificates for q=33 and q=39. Standard library only."""
import math
from fractions import Fraction
from functools import lru_cache

DATA = [
(33, '1/17', 93636,
 [1, 12, 12, -43, -43, 34, 34, -10, -10, 1, 1],
 '''
1000100222:861 1111112012:6966 2211020212:738 1020111021:3291
2021012210:9426 2110221112:492 1010122220:537 1110122002:279 0022122120:8934
0211221201:6355 2011120111:6545 2220010111:753 0211021122:246
1210210002:5028 1201020110:7506 1012011110:6099 1102211121:408
0011010102:12240 2120011011:1092 2010010001:4521 1201111112:4167
2002221222:408 0022221202:3437 2102221011:1830 1122111202:16 1100201122:1461
'''),
(39, '1/20', 131434191810,
 [1, -12, -12, 79, 79, -103, -103, 53, 53, -12, -12, 1, 1],
 '''
102112020210:1219154721 210001220021:628594164 010210102001:3382052226
002202222201:3675239955 010111122122:1728477428 021100020010:2940379970
100111000001:3965588986 200211122010:2906351488 010011102102:4734733751
111222202210:3564520628 020102000120:5735642133 100112021101:4034972285
221111021112:2788054669 020022212001:7937340702 202022002101:637385244
111122121020:8713871385 221201220112:3725747503 002210102000:5800572994
020110100222:1417269602 211222220111:3281813885 102222220002:9389547438
010021111110:1490195401 202011220110:10254004493 110100110021:725348383
222201122022:6875879692 112000100021:1660697195 210002210000:5226070479
000100102220:4969493108 110010001010:8367080149 221110110102:3745625330
212211012222:4187656967 122021112212:1724829456
'''),
]

def trim(a):
 while len(a)>1 and a[-1]==0:a.pop()
 return a

def divide(a,b):
 a=a[:];b=trim(b[:]);assert b[-1]==1
 quo=[0]*max(1,len(a)-len(b)+1)
 while len(a)>=len(b) and any(a):
  j=len(a)-len(b);v=a[-1];quo[j]=v
  for i,x in enumerate(b):a[i+j]-=v*x
  trim(a)
 return trim(quo),trim(a)

@lru_cache(None)
def cyclotomic(n):
 a=[-1]+[0]*(n-1)+[1]
 for d in range(1,n):
  if n%d==0:
   a,r=divide(a,list(cyclotomic(d)));assert r==[0]
 return tuple(a)

def verify(q,c_text,den,P,pairs):
 pairs=[s.split(':') for s in pairs.split()]
 seeds=[list(map(int,z)) for z,a in pairs]
 nums=[int(a) for z,a in pairs]
 assert q%2 and q>=3
```

```
D=len(P)-1;assert P[-1]==1
# Verify the real polynomial by its exact cyclotomic lift.
lift=[0]*(2*D+1)
for a,coef in enumerate(P):
 for b in range(a+1):lift[D+2*b-a]+=coef*math.comb(a,b)
assert tuple(trim(lift))==cyclotomic(2*q)
# The lift implies P(alpha)=0 and degree(P)=phi(2q)/2.
f=[[2],[0,1]]
for j in range(2,q):
 a=[0]+f[-1];b=f[-2]
 a += [0]*max(0,len(b)-len(a))
 for i,v in enumerate(b):a[i]-=v
 f.append(divide(trim(a),P)[1])
inds=list(range(1,q,2));pos={j:i for i,j in enumerate(inds)};m=len(inds)
V=[f[j]+[0]*(D-len(f[j])) for j in inds]
perms=[]
for a in range(1,q):
 if math.gcd(a,2*q)==1:
  pp=[]
  for j in inds:
   z=(a*j)%(2*q);z=min(z,2*q-z);pp.append(pos[z])
  perms.append(pp)
assert len(perms)==D
assert len(seeds)==len(nums) and all(a>0 for a in nums) and sum(nums)==den
assert all(len(z)==D and all(a in (0,1,2) for a in z) for z in seeds)
c=Fraction(c_text);assert c>Fraction(1,q-1)
# Accumulate original (not orbit-reduced) moments.
diff=[[0]*m for _ in range(m)];summ=[[0]*m for _ in range(m)]
for seed,num in zip(seeds,nums):
 e=[sum(a*b for a,b in zip(v,seed))%3 for v in V]
 for pp in perms:
  ee=[e[j] for j in pp]
  for i in range(m):
   for j in range(i,m):summ[i][j]+=num*(2 if (ee[i]+ee[j])%3==0 else -1)
   for j in range(i):diff[i][j]+=num*(2 if (ee[i]-ee[j])%3==0 else -1)
scale=2*D*den
for i in range(m):
 for j in range(i,m):assert summ[i][j]==0
 for j in range(i):assert Fraction(diff[i][j],scale)==-c
print(f'q={q}: EXACT PASS; {len(seeds)} seeds x {D} Galois images; c={c} > 1/{q-1}.')


for entry in DATA: verify(*entry)
```

# 4 Twice an odd prime power

The zero eigenvalue creates an additional first-moment constraint when $q$ is even. The following construction handles that constraint as well as the relations among the nonzero eigenvalues.

**Theorem 4.1.** *Let $p \geq 5$ be prime and $k \geq 2$. There is a bounded, compactly supported probability density on $[0, \infty)$ for which the averaged mixing matrix of $P_{2p^k-1}$ is exactly $J/(2p^k - 1)$.*

*Proof.* Set

$$s = p^k, \quad q = 2s, \quad h = (p-1)/2, \quad r = (p^{k-1} - 1)/2, \quad \ell = r + h, \quad c_0 = (2s-1)^{-1}.$$

For $c$ near $c_0$ define

$$B = \sqrt{p(1-(p-1)c)}, \quad \beta = B/p, \quad a = \frac{pc}{1-(p-1)c}; \qquad a\beta^2 = c. \tag{EP1}$$

At $c = c_0$, $a = (4r+1)^{-1}$. We shall construct the required coherences for $c > c_0$ sufficiently close, then use a positive Haar component and the exact finite-time transfer proved earlier.

**The two independent frequency sectors.** Write $\zeta = e^{\pi i/(2s)}$ and $\lambda_j = \zeta^j + \zeta^{-j}$, $1 \le j < 2s$. The field automorphism $\zeta \mapsto -\zeta$ fixes every even-index frequency and negates every odd-index frequency. It is indeed an automorphism, since $\gcd(2s+1, 4s) = 1$. Thus the rational spans of the two sectors intersect only at zero. The odd sector contains the zero mode $j = s$, whose phase is fixed at one.

After selecting one signed representative from each nonzero pair $\{\lambda, -\lambda\}$, each sector has $pr + h$ representatives, partitioned into $r$ disjoint $p$-tuples with signed sum zero and $h$ free representatives. Here is a direct construction for both sectors. Put $t = p^{k-1}$. In the even sector, write $\lambda_{2u} = 2\cos(\pi u/s)$ and choose the odd representatives $1 \le u \le s-2$. For each odd $1 \le j < t$, take the $p$ exponents $j + 2tv$, $0 \le v < p$, modulo $2s$, and reflect into $[0, s]$. They are distinct odd indices with cosine sum zero. Their residue classes modulo $2t$, up to sign, show that the blocks are disjoint; the unused representatives are $t, 3t, \dots, (p-2)t$. These representatives span the real field $\mathbb{Q}(2\cos(\pi/s))$, of degree $\varphi(s)/2 = ht$. Indeed their sum is 1, and every cosine at an integral multiple of $\pi/s$ is their signed representative or the constant 2; powers of $2\cos(\pi/s)$ expand into such cosines.

In the odd sector, for each odd $1 \le j < t$, sum the $p$ cosines with exponents $j + 4tu$, $0 \le u < p$, modulo $4s$. Their sum is zero. Reflect the exponents into $[0, 2s]$ and then into $[0, s]$, recording a minus sign at the second reflection. This gives $p$ distinct odd representatives. Residue classes modulo $2t$, up to sign, prove that the resulting tuples are disjoint. The unused representatives are $t, 3t, \dots, (p-2)t$.

The odd frequency span is the negative eigenspace of the displayed field involution. Odd powers of $2\cos(\pi/(2s))$ span that eigenspace and expand into odd cosines; consequently its dimension is $\varphi(s)/2 = ht$. The even sector has the same dimension. Since $pr + h - ht = r$, the displayed relations exhaust the rational relations in each sector. Their integer lattices are saturated: the rows have disjoint supports and coefficients $\pm 1$. Conjugating representative coordinates where necessary makes every block relation a product equal to one. There are no cross-sector relations. The time-flow closure is therefore the direct product of two tori, each consisting of $r$ product-one $p$-blocks and $h$ free phases. The character criterion also gives equidistribution of the actual nonnegative-time flow on this torus.

**An exchangeable root-label construction.** For either $\mu = 0$ or $\mu = -a\beta$, we construct $\ell$ exchangeable $p$th roots $Y_i$ satisfying

$$\mathbb{E}Y_i = \mu, \qquad \mathbb{E}Y_i^2 = -a, \qquad \mathbb{E}Y_i\overline{Y_j} = \mathbb{E}Y_iY_j = -a \quad (i \ne j). \tag{EP2}$$

Here and below all prescribed moments are real. We first treat $k \ge 3$. Since $\ell \equiv -1 \pmod p$, take $(\ell+1)/p$ complete sets of the $p$th roots, remove one copy of 1, and uniformly permute the remaining multiset. Its sum and its sum of squares both equal $-1$. Rotate it by a common $p$th root $Z$, then independently conjugate each coordinate with probability $1/2$. Put $z = a\ell$ and $b = -\ell\mu$, and assign to $Z = e^{i\theta}$ the mass

$$\frac{1 + 2b\cos\theta + 2z\cos(2\theta)}{p}. \tag{EP3}$$

Fourier orthogonality on the $p$th roots, using $p \ge 5$, gives $\mathbb{E}Z = b$ and $\mathbb{E}Z^2 = z$. The constructed vector has first moment $-b/\ell = \mu$, second moment $-z/\ell = -a$, and both off-diagonal moments equal $-d$, where

$$d = \frac{1}{2\ell} - \frac{a}{\ell - 1}. \tag{EP4}$$

Indeed, before the independent conjugations the sum of real parts is $-\cos\theta$. Its expected square is $(1+z)/2$, whereas the expected sum of squared real parts is $(\ell - z)/2$. Subtracting and dividing by $\ell(\ell-1)$ proves (EP4).

These are probability masses with a strict margin at $c_0$. Since $r \ge 2h(h+1)$ and $h \ge 2$,

$$z = \frac{r+h}{4r+1} \le \frac{2}{7}, \qquad \beta^2 < \frac{1}{p} \le \frac{1}{5}.$$

For $x = \cos\theta$, the numerator in (EP3) is at least

$$1 - 2z - \frac{b^2}{4z} \ge \frac{29}{70} > 0.$$

An iid root vector with single-coordinate masses

$$\frac{1 + 2\mu\cos\theta - 2a\cos(2\theta)}{p} \tag{EP5}$$

has the same first and second moments and off-diagonal moments $\mu^2$. Its masses are positive: at $c_0$, $a \le 1/49$ and $|\mu| \le a/\sqrt{5}$. Finally $d > a$, because this is equivalent to

$$(4r+1)(\ell-1) > 2\ell^2.$$

The difference is $2r^2 - 3r + h - 1 - 2h^2$; it increases with $r$ and, at $r = 2h(h+1)$, equals $8h^4 + 16h^3 - 5h - 1 > 0$. Mix the dependent law with the iid law, giving the former weight $(a+\mu^2)/(d+\mu^2) \in (0,1)$. This proves (EP2). All inequalities are strict, so the construction persists for $c > c_0$ sufficiently close.

**The square case.** It remains to construct (EP2) when $k = 2$, for which $r = h$ and $\ell = p - 1$. At $c_0$,

$$a = \frac{1}{2p-1}, \qquad \beta^2 = \frac{2p-1}{2p^2-1}.$$

Choose the integer $j$ nearest to $p/6$ and put

$$u = \cos(2\pi j/p), \quad \sigma = 1 - 2u, \quad \tau = 3 - 4u^2, \quad D = 1 + (\tau - \sigma^2)/2.$$

Since $p \equiv \pm 1 \pmod 6$, the angle lies in $[2\pi/7, 2\pi/5]$. Hence $3/10 < u < 5/8$ and

$$\sigma^2 < 4/25, \qquad \tau > 23/16, \qquad 27/16 < D < 9/4. \tag{EP7}$$

The upper bound for $u$ follows, for example, from the polynomial $8u^3 + 4u^2 - 4u - 1$ for $\cos(2\pi/7)$, which is increasing between $1/2$ and $5/8$ and positive at $5/8$.

Use two root multisets of length $\ell$. Let $M_0$ be all $p$th roots with 1 removed. Let $M_1$ be all $p$th roots with $1, e^{2\pi ij/p}, e^{-2\pi ij/p}$ removed and two copies of 1 added. Their sums and square sums are respectively $(-1,-1)$ and $(\sigma,\tau)$. Uniformly permute the chosen multiset, rotate it by a common root, then independently conjugate each coordinate. Define

$$V = \ell/2 - a\ell(\ell - 1/2), \quad w = \frac{1 - \ell + 2a\ell^2}{D}, \quad z_0 = \frac{a\ell - \tau w/2}{1-w}, \quad b_0 = \frac{-\ell\mu}{1-w}.$$

Choose $M_1$ with probability $w$, using rotation masses $(1-\cos(2\theta))/p$. Choose $M_0$ with probability $1 - w$, using rotation masses

$$\frac{1 + 2b_0\cos\theta + 2z_0\cos(2\theta)}{p}. \tag{EP8}$$

The first two total moments are

$$\mathbb{E}\sum_i Y_i = -(1-w)b_0 = \ell\mu, \qquad \mathbb{E}\sum_i Y_i^2 = -(1-w)z_0 - w\tau/2 = -a\ell.$$

Moreover, writing $C_i = \operatorname{Re} Y_i$, we have

$$\mathbb{E}\Big(\sum_i C_i\Big)^2 = \frac{(1-w)(1+z_0)}{2} + \frac{w\sigma^2}{4} = \frac{1+a\ell - wD}{2} = V, \qquad \mathbb{E}\sum_i C_i^2 = \frac{\ell(1-a)}{2}.$$

Exchangeability and the independent conjugations make both off-diagonal moments equal to $[V - \ell(1-a)/2]/[\ell(\ell-1)] = -a$. Thus (EP2) holds once these are verified to be probability laws.

At $c_0$ put

$$H = (2p-1)D - p, \qquad N = (p-1)D - p\tau/2.$$

Then $w = p/[(2p-1)D] \in (0,1)$, $z_0 = N/H$, and $|b_0| \le \beta(p-1)D/H$. Also

$$N = \frac{p(2-\sigma^2)}{2} - D > \frac{23p}{25} - \frac{9}{4} > 0, \qquad 1 - 2z_0 = \frac{D + p(\tau-1)}{H} > 0.$$

The numerator of (EP8) is bounded below on the whole circle by $1 - 2z_0 - b_0^2/(4z_0)$. This is positive provided

$$4N[D + p(\tau-1)] > \beta^2(p-1)^2D^2.$$

Since $\beta^2 < 1/p$, (EP7) reduces this to the sufficient inequality

$$4p\Big(\frac{23p}{25} - \frac{9}{4}\Big)(7p+27) > 81(p-1)^2.$$

The left side minus the right side is $644p^3/25 - 1116p^2/25 - 81p - 81$, which equals 1618 at $p = 5$ and has positive increasing derivative for $p \ge 5$. Therefore (EP8) has a strict positive margin. The other rotation law is nonnegative and fixed. All strict conditions persist for $c > c_0$ sufficiently close, completing (EP2) also for $k = 2$.

**Templates with two power sums.** There exist $p$ unit phases $v_1, \dots, v_p$ such that

$$\prod v_i = 1, \qquad \sum v_i = B, \qquad \sum v_i^2 = B^2/p. \tag{EP6}$$

Use a template $(1, e^{i\theta_1}, e^{-i\theta_1}, \dots, e^{i\theta_h}, e^{-i\theta_h})$. With $x_j = \cos\theta_j$, it suffices to prescribe

$$\sum x_j = (B-1)/2 =: S, \qquad \sum x_j^2 = h(1-c)/2 =: Q.$$

For the chosen $c$ we have $0 < c < 1/p$ and $1 < B < \sqrt{p}$. On the connected cube slice $\sum x_j = S$, the minimum square sum is $S^2/h < 1/2$, while $Q > h(1-1/p)/2 \ge 4/5$. An extreme point of this slice has at least $h-1$ coordinates in $\{-1, 1\}$, so its square sum is at least $h - 1 \ge h/2 > Q$. Continuity supplies the desired template.

**The phase law and the zero mode.** Use the first $r$ labels in (EP2) for the $p$-blocks. In block $a$, uniformly permute the template (EP6) multiplied by $Y_a$. The product is $Y_a^p = 1$, the conditional coordinate mean is $\beta Y_a$, and both the diagonal and distinct holomorphic second moments are $\beta^2 \mathbb{E} Y_a^2 = -c$. The distinct internal Hermitian moment is $(B^2 - p)/(p(p-1)) = -c$.

For each of the $h$ free phases, use one remaining label: retain it with probability $\beta$, and otherwise use an independent $p$th root of mean zero and second moment $a\beta$. The latter is supplied by masses $(1 + 2a\beta\cos(2\theta))/p$, which are positive. Its resulting first moment is $\beta\mu$, and its second moment is

$$\beta(-a) + (1-\beta)a\beta = -a\beta^2 = -c.$$

All distinct cross-block, block/free and free/free Hermitian and holomorphic moments are also $\beta^2(-a) = -c$. Thus every representative has first moment $\beta\mu$.

Choose $\mu = 0$ for the even sector and $\mu = -a\beta$ for the odd sector, and take the two sector laws independently. The first moments are respectively zero and $-c$. Every same-parity coherence between distinct modes is $-c$, including a mode and its negative partner, and including the fixed zero mode in the odd sector. Every opposite-parity coherence is zero by independence. Independent conjugations used in orienting the signed relations preserve all these real target moments.

We have constructed the full forced-coherence target at $c > c_0$ on the exact time-flow torus. Mixing with Haar measure in proportion $c_0/c$ gives the target at $c_0$ and a positive Haar component. All nonzero frequency-difference characters have Haar mean zero. The previously proved smoothing, finite-character correction and finite-time Gram correction yield an actual bounded compactly supported density on $[0, \infty)$. The path spectral identity now gives $J/(q-1)$. □

Together with the known positive result for twice a prime and the known parity-collision obstruction, this classifies all moduli $q = 2p^k$ with $p$ odd prime and $k \geq 1$: the negative cases are exactly $p = 3$, $k \geq 2$. The next section extends the positive construction to higher powers of two when $k \geq 2$. General mixed-prime moduli remain outside this theorem. This is a classification of one infinite family, not an all-path classification.

## 5 Higher powers of two times an odd prime power

The same templates also accommodate a larger power of two in the path modulus. The new ingredient is a root-label law for the changed numbers of relation blocks and free phases.

**Theorem 5.1.** *Let $p \geq 5$ be prime, $k \geq 2$ and $A \geq 2$. There is a bounded, compactly supported probability density on $[0, \infty)$ whose averaged mixing matrix on $P_{2^A p^k - 1}$ is exactly $J/(2^A p^k - 1)$.*

*Proof.* Put $R = 2^{A-2} p^{k-1}$, so $q = 4pR$, and use $B, \beta, a$ as in (EP1), now with $c_0 = (4pR-1)^{-1}$. At $c_0$,

$$a = (4R-1)^{-1}, \qquad \beta^2 = (4R-1)/(4pR-1) < 1/p.$$

In particular $p \mid R$ and $R \geq p$.

**The relation lattice.** Let $\zeta = e^{2\pi i/(8pR)}$. The involution $\zeta \mapsto -\zeta$ separates even and odd index frequency spans. Both have dimension $(p-1)R$. Indeed, the even span is the real field of the primitive $q$th root, of degree $\varphi(q)/2$; the constant belongs to this span by the sum of the nontrivial $p$th roots, and cosine expansions of powers give the whole field. The odd span is the negative eigenspace, since odd powers of $\zeta + \zeta^{-1}$ expand into odd-index cosines.

For each $1 \leq j < 2R$, sum the $p$ cosines with exponents $j + 8Rv$, $0 \leq v < p$, modulo $8pR$. Reflect into $[0, 4pR]$ and then $[0, 2pR]$, recording the sign of the second reflection. These are disjoint $p$-tuples: their residues modulo $4R$, up to sign, distinguish them, and $0 < j < 2R$ excludes a repeated or zero representative. Odd $j$ give $R$ blocks covering all $pR$ odd representatives. Even $j$ give $R-1$ blocks, leaving the $p-1$ free representatives $2R, 4R, \ldots, 2(p-1)R$. The zero mode is $j = 2pR$, which is even. The dimension counts give exactly $R$ and $R-1$ relations respectively. Disjoint support and coefficients $\pm 1$ make the displayed relation lattices saturated. Thus the exact time-flow torus consists of these product-one blocks and free phases, with independent parity sectors.

**Odd-sector labels.** Take $R/p$ complete sets of $p$th roots, uniformly permute them, and independently conjugate each coordinate. The first and diagonal holomorphic moments vanish,

while both off-diagonal moments are $-1/[2(R-1)]$. Mix this law with weight $w = 2a(R-1)$ and an iid law with masses

$$\frac{1 - 2a\cos(2\theta)/(1-w)}{p}.$$

The result has zero first moments and every required second moment $-a$. At $c_0$, $0 < w < 1$ and $a/(1-w) = 1/(2R+1) < 1/2$, so this remains valid for $c > c_0$ sufficiently close.

**Even-sector labels away from two exceptional cases.** Set $\ell = R + p - 2$ and $\mu = -a\beta$. Take $(\ell+2)/p$ full root sets, remove two copies of 1, uniformly permute, rotate by a common root of first moment $b = -\ell\mu/2$ and second moment $z = a\ell/2$, then independently conjugate the coordinates. The rotation masses are

$$\frac{1 + 2b\cos\theta + 2z\cos(2\theta)}{p}. \tag{HP1}$$

The multiset sum and square sum are both $-2$. Thus first moments are $\mu$, diagonal holomorphic moments are $-a$, and both off-diagonal moments are $-d$, with

$$d = \frac{(1-3a)\ell - 4}{2\ell(\ell-1)}.$$

For the last formula, the squared total real part has expectation $2(1+z)$ and the sum of squared real parts has expectation $(\ell - 2z)/2$. Mix this law with the iid law (EP5), using dependent weight $(a+\mu^2)/(d+\mu^2)$. This gives every second moment $-a$ whenever $d > a$.

At $c_0$, this inequality is equivalent to

$$F := R^2 - 9R - p^2 + 3p > 0. \tag{HP2}$$

For $R \geq 2p$, $F$ increases with $R$ and at $R = 2p$ equals $3p(p-5)$. Hence (HP2) holds except at $(p, R) = (5, 10)$. The allowed $R$ values have no value strictly between $p$ and $2p$. For $R \geq 2p$, $z < 3/16$ and $|b| = z\beta$, so the numerator of (HP1) is greater than $197/320$. Also $a \leq 1/39$ and $|\mu| < a/\sqrt{5}$, so (EP5) is strictly positive. All these conditions persist for $c > c_0$ close.

**The cases $R = p$ and $(p, R) = (5, 10)$.** Choose the integer $j$ nearest $p/4$ and put

$$u = \cos(2\pi j/p), \quad v = u^2 = \sin^2(\pi/(2p)), \quad T = 2 - 4v, \quad D = 3(1-v).$$

Since $\pi^2 < 10$, $0 < v < 5/(2p^2)$. Let $M_0$ be the preceding multiset with sum and square sum $-2$. From $M_0$, remove one copy of each of $e^{2\pi ij/p}, e^{-2\pi ij/p}$ and add two copies of 1, producing $M_1$ with sum $-2u$ and square sum $T$. Both multisets have $\ell$ entries. Define

$$V = \ell/2 - a\ell(\ell - 1/2), \quad w = \frac{2 + a\ell - V}{D}, \quad z_0 = \frac{a\ell - wT/2}{2(1-w)}, \quad b_0 = \frac{-\ell\mu}{2(1-w)}.$$

Use $M_1$ with probability $w$, rotating with masses $(1 - \cos(2\theta))/p$. Use $M_0$ with probability $1 - w$, rotating with masses

$$\frac{1 + 2b_0\cos\theta + 2z_0\cos(2\theta)}{p}. \tag{HP3}$$

Uniformly permute and independently conjugate in each case. The total first moment is $-2(1-w)b_0 = \ell\mu$, the total holomorphic square moment is $-2(1-w)z_0 - wT/2 = -a\ell$, and the squared total real part has expectation

$$2(1-w)(1+z_0) + wv = 2 + a\ell - wD = V.$$

Consequently both off-diagonal moments equal $-a$.

For $R = p$, let $H = (2p-1) - (4p-1)v$. At $c_0$ the parameters are

$$w = \frac{2p}{(4p-1)(1-v)} \in (0,1), \qquad z_0 = \frac{(p+1)v - 1}{H} < 0, \qquad |b_0| = \frac{(p-1)(1-v)\beta}{H}.$$

Here $H > 0$ and $z_0 < 0$ follow from $v < 5/(2p^2)$ and $p \geq 5$. The numerator of (HP3) is at least

$$1 + 2z_0 - 2|b_0| = \frac{(1-v)[2p - 3 - 2(p-1)\beta]}{H} > 0,$$

since $\beta < 1/\sqrt{p} < 1/2$. All necessary inequalities are strict and persist for $c > c_0$ close. For $(p, R) = (5, 10)$, (HP2) is an equality and $w(c_0) = 0$. At $c_0$, (HP3) is the strictly positive law (HP1). Since $dw/da = \ell(\ell + 1/2)/D > 0$, the same construction is valid with $0 < w < 1$ for $c > c_0$ sufficiently close.

**Assembly and actual time.** Apply template (EP6) to the blocks. For each even free phase, retain its label with probability $\beta$ and otherwise use independent root noise of first moment zero and second moment $a\beta$, exactly as in the previous theorem. All distinct same-parity coherences are $-c$, including the negative partners; first moments are zero in the odd sector and $-c$ in the even sector, giving the correct coherences with the fixed even zero mode. Independent sectors give zero opposite-parity coherences. Thus the full target holds at some $c > c_0$. The positive Haar mixture and exact finite-time transfer yield the claimed density. □

This extension leaves moduli $2^A p$ with $A \geq 2$ and general mixed odd-prime factors outside the construction. It is not an all-path classification.

## 6 What the constructions do and do not resolve

The local-potential problem is a finite auxiliary problem with an exact answer: the displayed incidence rules admit the rational certificate with constant 875/68. The profile constraints sharpen its consequence from twelve vertices to eleven. Applied to the original radius inequality, this proves the complete degree-seven case for graphs of arbitrary order. It does not remove the degree restriction. In particular, a proof of the auxiliary certificate problem is not being presented as a resolution of the unrestricted Fajtlowicz conjecture.

The phase-moment problem has two additional mechanisms: a quadratic torus density couples a dependent phase to the remaining modes, and exchangeable root multisets satisfy the first moments imposed by a fixed zero mode. Strict feasibility allows an exact correction along the actual time flow. These constructions settle the stated path families with the observation rule unchanged. A complete classification of all path moduli remains outside the results.

The source search checked the repository snapshot, the radius papers [1, 2], the quantum-walk problem survey [5], and Kang's arithmetic classification [6], together with targeted searches for higher-degree radius bounds and the path families treated here. No identical solution was identified in the inspected material. This is a bounded search, not a proof that no earlier equivalent argument exists. Neither a historically unprecedented principle nor a ranking above all previous corpus results follows from it. The manuscript is a research draft supported by internal independent checks; it has not undergone external peer review or proof-assistant verification.

## A Reproducibility from this single source

The TeX source compiles without external figures, bibliography files, or data files. It also contains, after the document terminator, a compressed text bundle with the following contents:

- the full rational graph certificate and its Python verifier;
- the C finite-core checker and an independent Python checker;
- exact quantum certificates for $q = 33, 39$ and their verifier;
- the recorded finite graph checks and generator provenance.

Compression is only a transport format: all mathematical data are ordinary decimal integers or rational strings. Python's standard library suffices to extract and verify the rational certificates.

Save the following as `extract.py`, then run `python3 extract.py paper.tex`, replacing `paper.tex` with the actual source filename. The script creates a fresh `repro` directory beside the source.

```
import base64, json, pathlib, sys, zlib
source = pathlib.Path(sys.argv[1])
s = source.read_text(encoding="utf-8")
begin = "%" + "BEGIN_EMBEDDED_REPRO_BUNDLE"
end = "%" + "END_EMBEDDED_REPRO_BUNDLE"
payload = s.split(begin, 1)[1].split(end, 1)[0]
encoded = "".join(line[1:].strip() for line in
                  payload.splitlines() if line.startswith("%"))
files = json.loads(zlib.decompress(base64.b64decode(encoded)))
out = source.parent / "repro"
out.mkdir(exist_ok=False)
for name, content in files.items():
    assert pathlib.Path(name).name == name
    (out / name).write_text(content, encoding="utf-8")
print("Extracted", len(files), "files to", out)
```

From the extracted directory, the two fast exact checks are:

```
python3 graph_verify_compact.py graph_degree7_compact_certificate.json
python3 quantum_compute_exceptional_verify.py
```

The first must report 45 types, 802 allowed pairs, 896 profiles, 895 profiles covered by the fourteen ordinary dual vectors, and residual lower bound 103/84. The second checks every required Hermitian and holomorphic moment for both exceptional quantum cases. Neither verification needs a linear optimization package.

To reproduce the exhaustive graph step, install the cited version of `geng` from nauty [7]. Compile the bundled C checker and run the following commands from the extracted directory:

```
cc -O3 -std=c11 graph_corechk.c -o graph_corechk
for n in 4 5 6 7 8 9; do
    geng -C -d2 -q "$n" | ./graph_corechk
done
geng -C -d2 -D7 -q 10 | ./graph_corechk
geng -C -d2 -D7 -q 11 | ./graph_corechk
python3 graph_corecheck_python.py /absolute/path/to/geng
```

The C program's pass/fail decision uses only integers; its decimal minimum-slack display is diagnostic. The independent Python program uses exact fractions and checks orders four through nine. The two higher-order minimum-slack witnesses were also checked with exact fractions. The generator supplies completeness up to graph isomorphism; the checker does not itself certify the generator. The C parser is intended for these small, header-free graph6 records only.

The $q = 33, 39$ verifier is printed in full in Section 3.6. The remaining quantum proofs are analytic and require no finite search. Their use of continuity is justified by explicit strict inequalities at the target parameter; in the single boundary case $q = 200$, the mixture weight increases from zero on the required side while the other probability masses remain positive.

# B The global graph potential

For direct inspection, the first graph certificate is printed below. Each row gives $t = (d, p, l)$ and the multipliers in (7)–(9), with $C = 875/68$. The profile certificates and residual vector are in the embedded bundle. Their exact coefficient inequalities are specified in Lemma 2.6 and checked by the bundled program.

| $t$ | $\alpha_t$ | $\beta_t$ | $t$ | $\alpha_t$ | $\beta_t$ | $t$ | $\alpha_t$ | $\beta_t$ |
|---|---|---|---|---|---|---|---|---|
| $(3,1,1)$ | $0$ | $-\frac{671}{408}$ | $(5,1,2)$ | $\frac{61}{136}$ | $-\frac{125}{272}$ | $(6,1,6)$ | $\frac{1}{5}$ | $-\frac{137}{170}$ |
| $(4,0,1)$ | $\frac{875}{272}$ | $-\frac{201}{272}$ | $(5,1,3)$ | $\frac{313}{400}$ | $-\frac{787}{850}$ | $(7,0,1)$ | $\frac{137}{170}$ | $\frac{11}{340}$ |
| $(4,0,2)$ | $\frac{919}{544}$ | $-\frac{647}{544}$ | $(5,1,4)$ | $\frac{201}{272}$ | $-\frac{253}{136}$ | $(7,0,2)$ | $\frac{57}{136}$ | $\frac{11}{340}$ |
| $(4,0,3)$ | $\frac{671}{408}$ | $-\frac{535}{136}$ | $(5,1,5)$ | $\frac{1}{4}$ | $-\frac{253}{136}$ | $(7,0,3)$ | $\frac{74}{255}$ | $\frac{11}{340}$ |
| $(4,0,4)$ | $\frac{1}{4}$ | $-\frac{2743}{816}$ | $(6,0,1)$ | $\frac{253}{136}$ | $-\frac{117}{680}$ | $(7,0,4)$ | $\frac{373}{1360}$ | $-\frac{11}{340}$ |
| $(4,1,1)$ | $0$ | $-\frac{201}{272}$ | $(6,0,2)$ | $\frac{787}{850}$ | $-\frac{181}{850}$ | $(7,0,5)$ | $\frac{181}{850}$ | $-\frac{11}{340}$ |
| $(4,1,2)$ | $\frac{1007}{816}$ | $-\frac{647}{544}$ | $(6,0,3)$ | $\frac{53}{85}$ | $-\frac{74}{255}$ | $(7,0,6)$ | $\frac{117}{680}$ | $-\frac{11}{340}$ |
| $(4,1,3)$ | $\frac{671}{408}$ | $-\frac{2743}{816}$ | $(6,0,4)$ | $\frac{125}{272}$ | $-\frac{57}{136}$ | $(7,0,7)$ | $\frac{1}{7}$ | $-\frac{11}{340}$ |
| $(4,1,4)$ | $\frac{1}{3}$ | $-\frac{2743}{816}$ | $(6,0,5)$ | $\frac{107}{272}$ | $-\frac{263}{272}$ | $(7,1,1)$ | $0$ | $-\frac{19}{136}$ |
| $(5,0,1)$ | $\frac{175}{68}$ | $-\frac{107}{272}$ | $(6,0,6)$ | $\frac{1}{6}$ | $-\frac{137}{170}$ | $(7,1,2)$ | $\frac{1}{6}$ | $-\frac{19}{136}$ |
| $(5,0,2)$ | $\frac{647}{544}$ | $-\frac{125}{272}$ | $(6,1,1)$ | $0$ | $-\frac{467}{2040}$ | $(7,1,3)$ | $\frac{79}{408}$ | $-\frac{125}{816}$ |
| $(5,0,3)$ | $\frac{404}{425}$ | $-\frac{787}{850}$ | $(6,1,2)$ | $\frac{787}{850}$ | $-\frac{2093}{5100}$ | $(7,1,4)$ | $\frac{49}{272}$ | $-\frac{125}{816}$ |
| $(5,0,4)$ | $\frac{389}{544}$ | $-\frac{253}{136}$ | $(6,1,3)$ | $\frac{787}{850}$ | $-\frac{7271}{10200}$ | $(7,1,5)$ | $\frac{49}{272}$ | $-\frac{1}{6}$ |
| $(5,0,5)$ | $\frac{1}{5}$ | $-\frac{253}{136}$ | $(6,1,4)$ | $\frac{125}{272}$ | $-\frac{1009}{1632}$ | $(7,1,6)$ | $\frac{117}{680}$ | $-\frac{1}{6}$ |
| $(5,1,1)$ | $\frac{2743}{816}$ | $-\frac{107}{272}$ | $(6,1,5)$ | $\frac{107}{272}$ | $-\frac{409}{408}$ | $(7,1,7)$ | $\frac{1}{6}$ | $-\frac{1}{6}$ |

# References


[1] M. Cygan, M. Pilipczuk, and R. Škrekovski, *On the inequality between radius and Randić index for graphs*, MATCH Communications in Mathematical and in Computer Chemistry **67** (2012), 451–466. https://match.pmf.kg.ac.rs/electronic_versions/Match67/n2/match67n2_451-466.pdf.

[2] M. I. Doig, *Block eccentricity, a radius bound, and an application to the Randić index*, arXiv:2309.11613v1, 20 September 2023. https://arxiv.org/abs/2309.11613.

[3] QEEDDD, *Verifiable Mathematical Reasoning Corpus*, 102 numbered research drafts, snapshot `ecd87de3348dcb706a76bb12fde722d7a348a012`, 2 November 2026. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus.

[4] VMRC, item [087], *Spanning trees of small diameter and the Randić index*, file `[087] fajtlowicz_radius_core_reduction.tex`, unrefereed research draft in the snapshot of [3].

[5] G. Coutinho and K. Guo, *Selected open problems in continuous-time quantum walks*, Special Matrices **12** (2024), 20240025. doi:10.1515/spma-2024-0025.

[6] M. Kang, *Arithmetic of Bohr Frequencies Governs Uniform Mixing in Randomly Timed Quantum Spin Chains*, arXiv:2609.14463v1, 13 September 2026. https://arxiv.org/abs/2609.14463v1.

[7] B. D. McKay and A. Piperno, *Practical Graph Isomorphism, II*, Journal of Symbolic Computation **60** (2014), 94–112. doi:10.1016/j.jsc.2013.09.003. Software: https://users.cecs.anu.edu.au/~bdm/nauty/. The runs used nauty 2.9.0, source mirror and commit given in Section 2.